

Title: Multi-Agent System for Portfolio Optimization

Abstract: In the modern financial landscape, portfolio optimization is a critical aspect of maximizing returns while mitigating risks. To address this challenge, we propose a multi-agent system (MAS) architecture leveraging advanced AI models and tools to provide users with an optimized portfolio strategy. This article outlines the conceptual design, implementation, and potential benefits of the system, focusing on real-time data analysis, web-based insights, financial fundamentals, and news summarization.

Introduction: Portfolio management involves juggling various assets to achieve a balanced and profitable mix. However, the dynamic nature of markets, coupled with the vast amount of data available, poses significant challenges for investors. Our multi-agent system addresses these challenges by automating and integrating data collection, analysis, and summarization processes to provide a comprehensive portfolio report. The system utilizes cutting-edge AI models such as Gemini-Pro, GPT-3.5-Turbo, and Llama3-Groq-70B, alongside tools like OpenBB, DuckDuckGo, and YFinance.

System Architecture: The system comprises four key agents that work collaboratively to generate an optimized portfolio report. Below is an overview of the agents and their respective roles:

1. Real-Time Stock Data Agent:

- **Objective:** Fetch real-time stock data for portfolio items.
- **Model:** Llama3-Groq-70B-8192-Tool-Use-Preview.
- **Tools:** OpenBBTools.
- **Output:** Organized tabular data showcasing the latest stock prices and relevant metrics.

2. Web Search Agent:

- **Objective:** Gather additional stock-related information from credible web sources.
- **Model:** GPT-3.5-Turbo.
- **Tools:** DuckDuckGo.
- **Output:** Insights and links to credible web resources for further exploration.

3. Finance Data Agent:

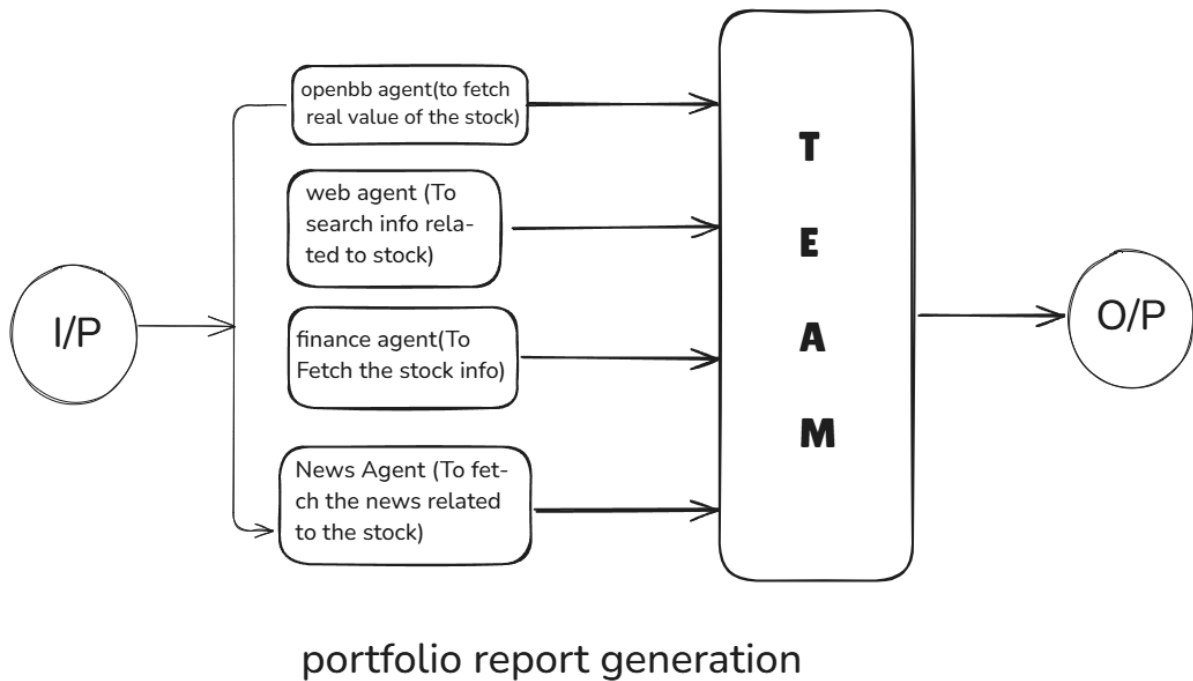
- **Objective:** Retrieve stock fundamentals, current prices, and analyst recommendations.
- **Model:** Gemini-Pro.
- **Tools:** YFinanceTools, CsvTools.
- **Output:** Clear tables presenting stock fundamentals, prices, and analyst ratings.

4. News Summarization Agent:

- **Objective:** Summarize recent news articles related to portfolio stocks.
- **Model:** Llama3-Groq-70B-8192-Tool-Use-Preview.
- **Tools:** Newspaper4k.
- **Output:** Concise news summaries with links to original articles.

Multi-Agent Coordination: To streamline the interaction between agents, a team-based approach is implemented. The **Portfolio Team** agent ensures efficient communication and coordination between the four individual agents. Its responsibilities include:

- Orchestrating the data retrieval and processing workflow.
- Integrating outputs from all agents into a cohesive report.
- Providing actionable insights to the user.



Models Utilized:

1. Gemini-Pro:

- A versatile multimodal language model capable of processing text, images, audio, and video.
- Offers computational efficiency and advanced reasoning capabilities.

2. GPT-3.5-Turbo:

- Provides enhanced accuracy and efficiency for language-based tasks.
- Generates up to 4,096 tokens, ideal for detailed portfolio analysis.

3. Llama3-Groq-70B:

- Specialized for tool-based integrations and handling complex queries.
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Implementation: The system starts by importing portfolio data from a CSV file. The **Real-Time Stock Data Agent** retrieves live market data using OpenBBTools, while the **Web Search Agent** supplements this with additional insights. The **Finance Data Agent** provides in-depth financial analysis, and the **News Summarization Agent** highlights recent developments affecting portfolio assets. Finally, the **Portfolio Team** synthesizes these outputs into a comprehensive report.

Potential Applications:

- **Retail Investors:** Automates portfolio analysis for individuals, enabling informed decision-making.
 - **Financial Institutions:** Provides detailed reports to support institutional investment strategies.
 - **Education:** Serves as a teaching tool for students learning about finance and portfolio management.
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Conclusion: By leveraging a multi-agent system with state-of-the-art AI models and tools, we can significantly enhance portfolio management processes. This innovative approach reduces the complexity of data analysis and empowers users with actionable insights. The system represents a step forward in financial technology, demonstrating the power of AI in solving real-world challenges.

Future Work: Future iterations of this system could incorporate:

- Enhanced visualization tools for report generation.
- Predictive analytics to forecast stock trends.
- Integration with blockchain for secure and transparent data handling.

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GitHub Repository: For implementation details, visit the GitHub repository: [Portfolio Optimization GitHub](#)